

Computer Architecture

CSIT — 3rd Semester — 2080 — Answer Sheet
Subject Code: CSC165

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

- 1. Explain the working of Booth's multiplication algorithm and perform multiplication of 50 and (-13) using the same algorithm.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 2. Explain the instruction format of basic computer. Write down symbolic microinstruction for FETCH routine.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 3. What is DMA? Explain the DMA controller with block diagram. How the DMA interact with I/O device.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

- 4. What is arithmetic overflow? How it can be detected?**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 5. What do you mean by error correction codes? Explain any one of error correction technique.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 6. Explain block diagram of microprogram sequencer in brief.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 7. What is cache memory? Explain the elements of cache design.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is micro-operation? Explain different arithmetic microoperations.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What do you mean by pipelining concept? Discuss various pipeline hazards and their solutions in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**10. Write the program for following statement by using three, single, zero address instructions.
 $X = (A + B * C - D) / (E * F + G)$**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Differentiate between RISC and CISC architecture.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on (Any TWO) a. Control ROM b. Common Bus System c. Flynn's Classification

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.